KOTAANA

How the app thinks

The workout algorithm, the nutrition algorithm, the dashboard maths and every API that connects them — explained step by step, with diagrams, formulas and worked examples.

Workout selection | Nutrition & AI meal plans | Health rules | Dashboard scores | Coach & payments | API map

Built from the application source · 12 October 2026 · Every number in this guide is read from the code, not estimated.

START HERE

How to read this guide

Kotaana is made of three layers. Once you see which layer does what, every feature becomes easy to follow.

1

Screens

What you tap: dashboard, workout, nutrition, coach portal. Screens never decide anything — they ask the server and show the answer.

2

APIs

Small server addresses such as `/api/workout/generate`. Each one checks who you are, loads your data from the database, calls an engine, and saves the result.

3

Engines

The rules and maths: the workout engine, the nutrition engine, the health rules and the AI router. They hold the real "intelligence".

Words used in this guide

Word	Meaning in Kotaana
Dataset / library	The stored list of real exercises (1,002) and real foods (about 8,100 from USDA). The app only recommends things that exist here.
Verified	A flag on each exercise/food. The workout engine ignores anything not verified.
Filter (safety gate)	A hard yes/no test. If an exercise fails it, it is removed for that person — it can never come back through scoring.
Score	A number used to rank what is left (higher = better fit). See page 8 for how it works today.
Seed	A number made from your user ID + today's date. It makes the daily pick different each day but stable if you reload.
Readiness	Your 0–100 daily score from the check-in (energy, mood, soreness, sleep).
AI (DeepSeek / Gemini)	A language model called over the internet. It suggests; the app's own rules always have the last word.
Fallback	What the app does when something is missing (no AI key, no weather, no data) — it never invents numbers.

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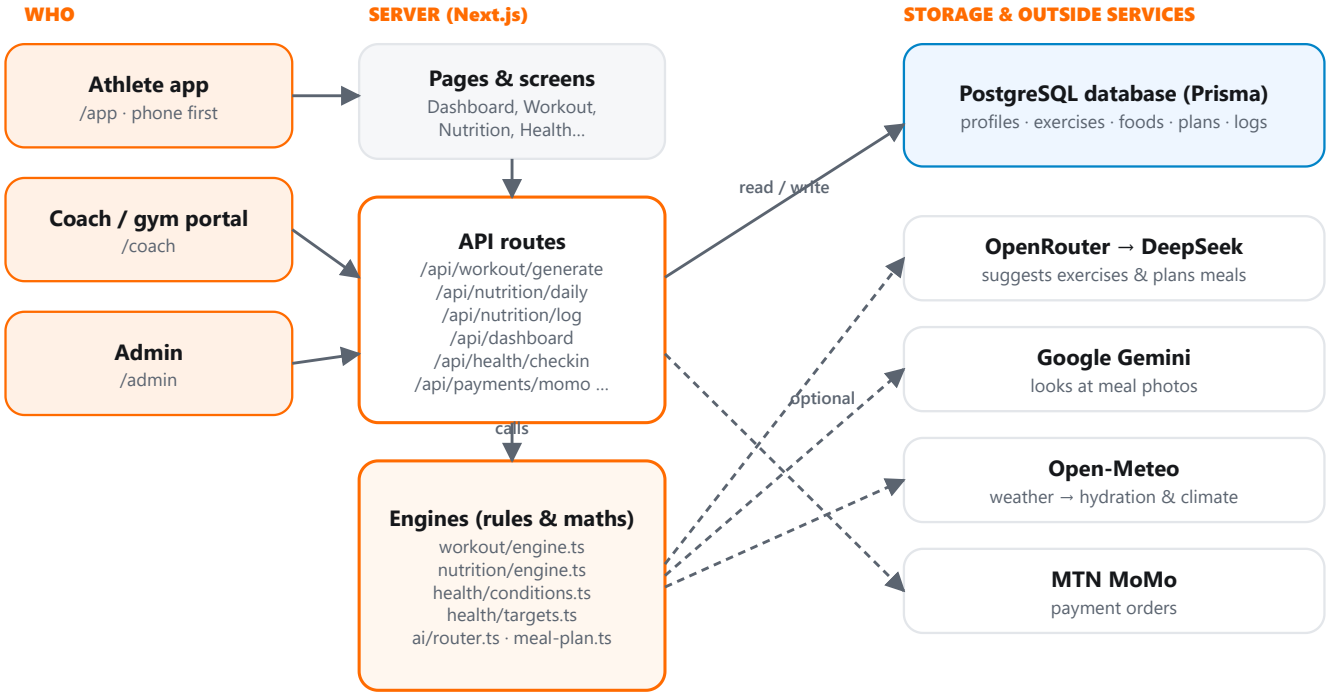
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PART A · SYSTEM

Architecture: who talks to whom

Every action travels the same road: screen → API route → engine → database, with outside services used only for extras.



Solid arrows = always. Dashed arrows = only when a key / network is available; otherwise the app falls back to its own rules.

The rule that keeps it safe

- **Screens never trust themselves.** Every API first calls `apiAuth(role)`: it reads your session and refuses the request if your role (athlete / coach / admin) is wrong.
- **Every row has an owner.** Reads are filtered by your user ID (or, for coaches, by an *active* coach–athlete link). That is how data reaches the right coach and nobody else.
- **AI is optional.** If no AI key exists, workouts and meals are still built by the app's own rules. If the AI answer is unsafe or unusable, it is thrown away.

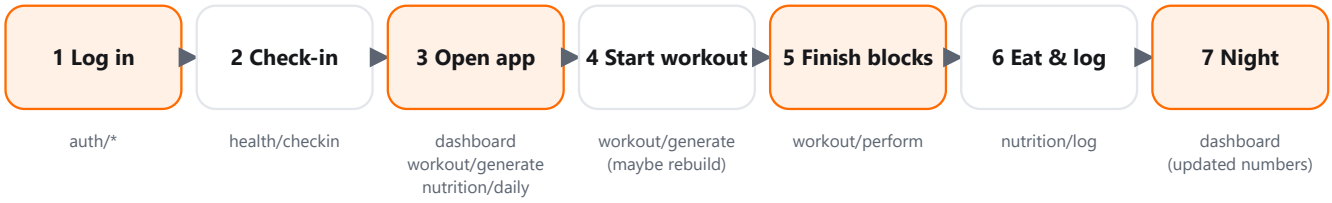
API groups

Group	Routes	Used for
Workout	workout/generate · workout/perform · exercises/[id]	Build today's session, record how each block went, show how-to demos
Nutrition	nutrition/daily · nutrition/log · hydration	Plan meals, log/verify them, count water
Health	health/checkin · health/injury · health/document · ai/risk	Readiness, injuries, documents, risk signal
Dashboard & reports	dashboard · reports/overview · reports/series · ai/weekly-summary · ai/monthly-progress	Numbers and charts, from recorded data only
Coach & money	connections/search · connections/request · coach/* · payments/momo · payments/momo/approve	Pick a coach, share data, pay and confirm
Account	auth/* · profile · location · notifications	Sign-in (email or Google), profile, GPS, inbox

PART A · SYSTEM

A day in the life of one athlete

Follow one person through a normal day and watch which APIs and engines fire.



Green text under each step = the API routes it touches.

Step	What you do	What the server does	Saved in
1	Sign in with email or Google	Checks password (bcrypt) or Google token, makes a session, sends you to onboarding or the dashboard	users, accounts, profiles
2	Answer the check-in card	Computes readiness: **50 + energy×8 + mood×5 − soreness×9 + min(12, sleep)×2**, clamped 0–100. Example (Okay = 3, 2, 3 and 7 h sleep): 50+24+15−18+14 = **85**	daily_checkins
3	Open the dashboard	Three parallel calls: workout/generate (GET) finds or builds today's session; nutrition/daily (GET) finds or builds today's meals; dashboard collects all numbers	workout_plans, nutrition_plans (cache)
4	Press Start Workout	First start of the day shows 4 questions (indoor/outdoor, how hard, aim, minutes). If answers changed, they are saved to your profile and the session is rebuilt	profiles, workout_plans (old plan switched off, new version saved)
5	Tap easy / ok / hard / skip after each block	workout/perform stores effort, updates the plan's completed list and the day's completion rate, and adds an energy estimate	exercise_performances, workout_logs, activity_data
6	Tap “I ate this” or take a photo	nutrition/log adds the meal (calories & macros) to today's log; with a photo Gemini compares plate and plan	meal_logs
7	Look at the dashboard	dashboard recalculates rings, streak, health overview and charts from everything saved above	(read only)

Why the plan can change by itself: when you open the app, the workout API compares timestamps. If you checked in or reported an injury *after* today's plan was made, it silently builds a new version (reason “health_change”). If there is no plan yet today, it builds one (reason “daily_auto”). Pressing Update in the questions card builds one too (”manual_generate”).

PART B · WORKOUT ALGORITHM

Step 1 — What the engine knows about you

Before choosing anything, `workout/generate` gathers everything relevant. Nothing is guessed: if a value is empty the engine uses a stated default and records it.

Input	Comes from	How it is used
Goal, track, level, minutes	profiles (+ the 4 questions)	Session length, number of exercises, sets/reps, bonus points. "How hard" shifts level: easy −1, hard +1 (beginner ↔ intermediate ↔ advanced)
Equipment you have / excluded	profiles + your gym's equipment list (if you have a coach)	Hard filter: an exercise needing equipment you lack is removed
Age band	profiles (from date of birth)	Hard filter for youth / restricted exercises
Injuries & limitations	injuries table (active/recovering) + profile notes	Hard filter: knee, shoulder, back, wrist, ankle rules (page 6)
Health conditions, pregnancy, stress	profiles	Removes exercise types, caps effort, adds notes (page 7). Stress ≥ 4 lowers readiness by 10
Readiness	today's daily_checkins (default 70 if none)	Lowers effort and volume when low
Last 14 days of performances	exercise_performances	Recent exercises/muscles/patterns; last effort on each exercise → progression
This week's muscles & patterns	exercise_performances (7 days)	Variety scoring
Missed sessions	workout_logs (7 days) vs your weekly target	Adds a note: volume held conservative if 2+ missed
Training days, outdoor OK	profiles	Picks one outdoor day per week (seeded)
Seed	hash(user ID + today's date)	Daily rotation that stays stable on reload

Where the exercises come from

Dataset	Records
free-exercise-db	745
bryllim / workout-guide	256
mohamedatef	1
Total in the library	**1,002**

Each record has: name, body part, target muscle, secondary muscles, equipment, difficulty, steps, GIF/image, and safety tags (contraindications, age limits, role tags such as *stretch*, *warmup*, *core*, *outdoor*).

Honest data-quality note. The curation report marks all 1,002 records **"partial"** quality and many lack written instructions with "human safety review required before activation". The engine therefore filters by tags and keywords. A person reviewing the tags will make every filter on the next pages more reliable.

PART B · WORKOUT ALGORITHM

Step 2 — The safety funnel

Hard yes/no gates run first. An exercise that fails any gate is gone for that person today, whatever its score would have been.



Shape only — the real size of each step differs for every person. The app shows the final count as "candidateCount".

Gate	Exact rule in the code	Example
Verified	exercise.verified must not be false	Unreviewed imports are ignored
Equipment	"body only / none / bodyweight" always passes. Otherwise the exercise's equipment text must match something you own. Anything in your "not available" list is removed	You own only dumbbells → barbell exercises vanish
Difficulty	exercise difficulty rank (1–3) ≤ your level rank + 1	A beginner can see beginner and intermediate, never advanced
Age	Youth bands drop exercises tagged adult_only / high_risk. Unknown age drops age_restricted / high_risk	Under-17 → no heavy loaded lifts
Injury	If injury status is unknown, drop high_risk / injury_sensitive / advanced_load tags. If known, each injury maps body part → forbidden tags & words (below)	Knee injury → jump, pistol, deep squat, box jump removed
Limitations	"no jumping / no running / no impact" in your notes removes jump, run, sprint, impact	
Health condition	Words listed by each condition (page 7) are searched in the exercise name, muscles, tags	Osteoporosis → crunch, twist, jump removed

Injury map (body part → what is removed)

Injury text contains	Contraindication tags	Words removed
knee	knee, squat_depth, impact	jump, pistol, deep squat, lunge jump, box jump
shoulder / rotator	shoulder, overhead	overhead press, snatch, handstand, dip, muscle up
back / lumbar	lower back, hinge	good morning, heavy deadlift, sit up, hanging leg raise
wrist	wrist, loaded wrist	handstand, planche, push up
ankle	ankle, impact	jump, sprint, box jump

If **nothing** survives the funnel, the engine stops with an error that lists what blocked it (equipment, age, level, injury…). It will not build an unsafe fallback — that is by design.

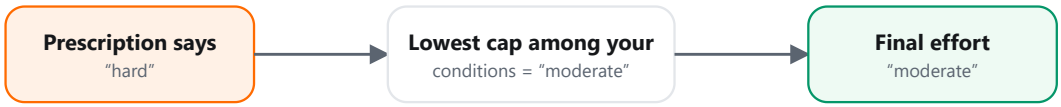
PART B · WORKOUT ALGORITHM

Step 3 — Health conditions change the rules

Each condition you tick in Health adds its own rule set. The same rules also remove foods (see the nutrition part).

Condition	Exercises removed	Effort cap	Foods removed
High blood pressure	sprint, burpee, box/jump/tuck jump, plyo, handstand, headstand	moderate	salty / processed
Heart disease	same as above	moderate	salty / processed
Diabetes (type 1, 2)	—	—	sugary foods
Asthma	sprint	—	—
Arthritis · Obesity	high-impact (jump, sprint, burpee, plyo, skipping)	—	—
Osteoporosis	high-impact + sit-ups, crunches, toe touch, twists	—	—
Chronic back pain	crunches, sit-ups, deadlift, hanging leg raise, superman	—	—
Sickle cell	high-intensity moves	moderate	+500 ml water
Epilepsy	handstand, headstand, rope climb	—	—
Kidney disease	—	moderate	salty foods
Pregnancy	high-impact, spinal flexion, twists, hanging, inversions	moderate	raw fish, liver, high-mercury fish, alcohol

How an effort cap works



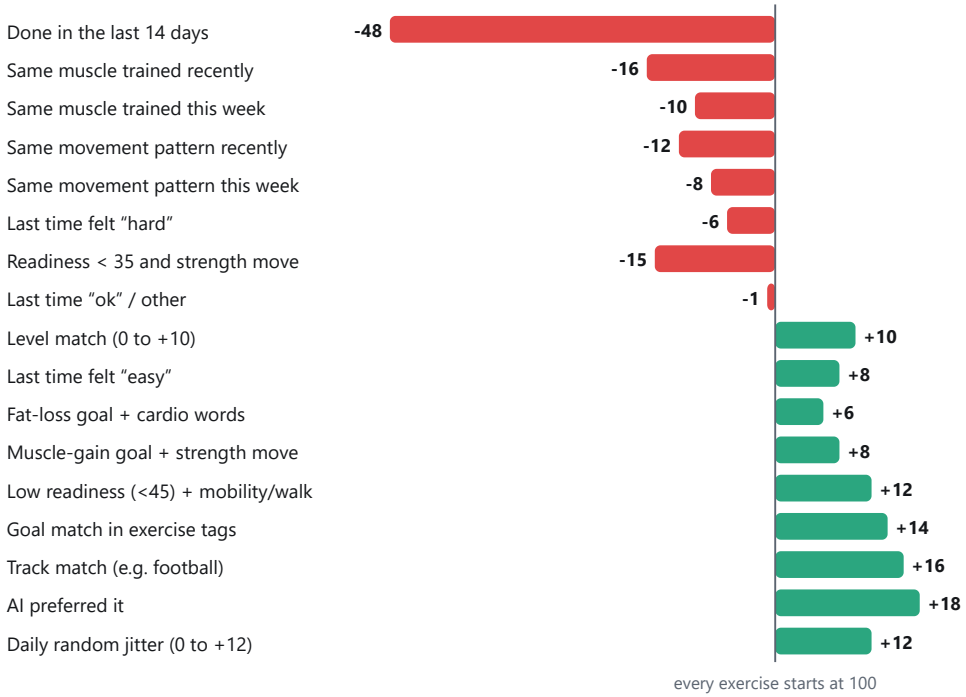
Caps only ever lower effort, never raise it.

Every active condition also adds a plain-language note that appears on the workout ("Adjusted for your health profile") and a nutrition note on the meals page. These are conservative filters and reminders — they never replace a doctor's clearance.

PART B · WORKOUT ALGORITHM

Step 4 — Scoring: how “good for you today” is calculated

After the funnel, each remaining exercise gets a score that starts at 100 and moves up or down. Blocked exercises get −∞ (never chosen).



Signal	Why it exists
Recent use, muscles, patterns	Variety and recovery: don't repeat the same work two days in a row
Goal / track match	Fat loss favours cardio, muscle gain favours strength moves, football track favours football-tagged moves
Readiness	Tired days favour mobility and walking and penalise heavy strength work
Last performance	Moves that felt easy are more welcome; moves that felt hard are less so
AI preferred	A DeepSeek suggestion adds +18 (page 11)

Important — read page 12 next. These scores are computed and sorted correctly, but in the current code the final pick does not use the score order. This page explains the *intended* ranking; page 12 shows what actually decides.

PART B · WORKOUT ALGORITHM

Step 5 — Building the session

The eligible exercises are sorted into role pools, then one session is assembled in a fixed order.

1. Role pools

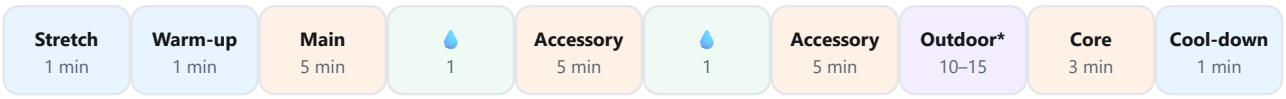
Pool	An exercise belongs when its tags/words say…
Stretch	role “stretch”, or stretch / mobility / flexibility / child pose / cat cow / 90 90
Warm-up	role “warmup”, or warm up / dynamic / march / jog / jumping jack / arm circle / leg swing
Core	role “core”, or core / abs / plank / dead bug / bird dog / oblique
Outdoor	role “outdoor”, or run / jog / walk / hike / sprint / hill
Main	everything left over (strength, cardio, bodyweight work)

If the stretch, warm-up or main pool is empty the engine stops and tells you which pool is missing — it will not make up a session.

2. How many main blocks

Your minutes	Main + accessory blocks
50 or more	5
38–49	4
28–37	3
under 28	2

3. Order of the session



* only on your seeded outdoor day each week, when outdoor training is allowed and outdoor exercises passed the filters

Shown for 35 min; the number of main/accessory blocks changes with your minutes.

4. Keep it inside your time

Minutes are added up from each block's estimate. While the total is more than **your minutes + 5**, the engine removes accessory blocks one by one (never the first main block).

5. Save

The old plan for today is switched off (`isActive=false`), the new plan is saved with a version number and a reason, and the screen shows it. The session carries a stamp of what the engine used: defaults applied, candidate count, adjustment note and the policy line.

PART B · WORKOUT ALGORITHM

Step 6 — Sets, reps and progression

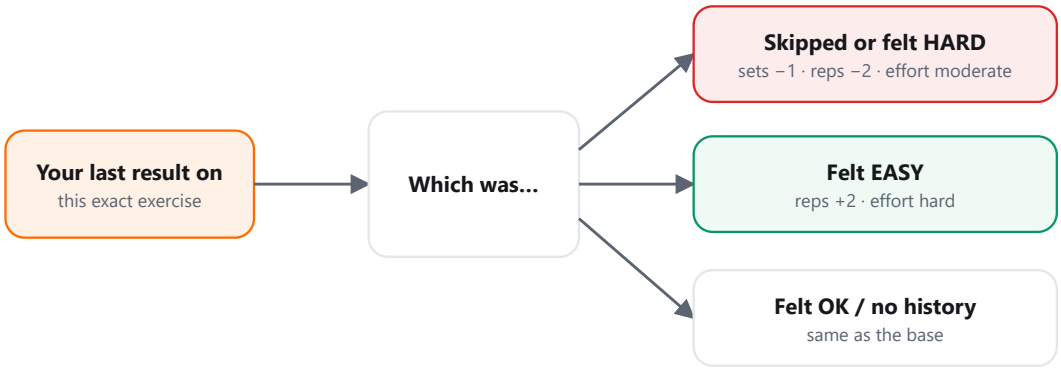
Choosing the exercise and deciding the dose are two separate jobs. The dose reacts to how your last attempt went.

Base dose by role

Role	Beginner	Intermediate	Advanced
Main / accessory	3 × 10 reps	3 × 12 reps	4 × 8–12 reps
Core	2 × 25 s	3 × 35 s	3 × 35 s
Outdoor	10 min	15 min	15 min
Stretch · warm-up · cool-down	45–60 s each, short rest	same	same

Rest is 60 s between main sets, 30 s for core. Minimum 2 sets and 6 reps after adjustments.

Progression from your last result



Readiness changes the dose

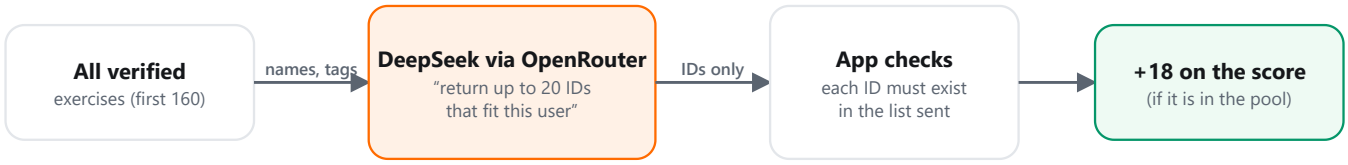
Readiness	Effect on the dose
≥ 45	Dose as above
< 45	Core holds are 25 s and easy; outdoor work is easy
< 35	Main sets −1 (minimum 2) and effort set to easy

Missed sessions: 2+ missed this week → note "volume held conservative before rebuilding frequency"; 1 missed → "normal volume with conservative progression". This is a message about the plan's intent; the dose itself is governed by the table above.

PART B · WORKOUT ALGORITHM

Step 7 — What the AI does (and cannot do)

The AI is a suggestion box with strict rules. It never sees medical data as a decision-maker and cannot add exercises.



Question	Answer
When is it called?	On every workout build, only if an AI key exists (OPENROUTER_API_KEY, else GEMINI_API_KEY)
What does it receive?	Age band, level, track, goal, minutes, readiness, and for up to 160 exercises: id, name, equipment, category, muscles, tags, difficulty
What can it return?	A JSON list of at most 20 exercise IDs. Anything not in the list it was given is discarded
System instruction	"You may only choose from verified IDs… Never invent exercise IDs, medical diagnoses, or unsafe instructions."
Can it override safety?	No. Filters run on all exercises regardless of the AI. The AI line only adds points to exercises that already passed
If the call fails?	The workout is built without it — nothing breaks
Which model?	OPENROUTER_MODEL, default deepseek/deepseek-chat-v3.1 (temperature 0.1)

PART B · WORKOUT ALGORITHM

Step 8 — Reality check: what actually decides the exercises today

Reading the code line by line, then testing it, shows the scores are not what picks the main exercises.

What the code does

1. Eligible exercises are scored and sorted best-first into pools.
2. `choose(pool, count, seed)` then **re-sorts the whole pool** by a pseudo-random key — `(seed + JSON length of the exercise × 31) mod 997` — and takes the first *count*.
3. The score order is therefore thrown away for stretch, warm-up, main and core picks.

A test that proves it

40 identical beginner exercises, same user and date:

Case	Main blocks chosen
Plain	e0, e7, e1
AI prefers e5, e6, e7 (+18 each)	e7, e0, e1 (e5, e6 not chosen)
20 exercises marked "done recently" (−48)	e0, e7, e1 (unchanged)

What this means in plain words

- **Safety is solid:** the filters work before choosing, so unsafe exercises cannot be picked.
- **Daily variety works:** the seed changes every day, so you get a different pick each day, the same all day.
- **Not working as designed:** "avoid what I did yesterday", "favour my goal/track", "prefer easier moves when tired" and the AI's +18 are calculated but do not steer the pick. Today's choice is a seeded random draw from everything that passed the safety funnel.
- **Still working:** sets, reps, effort and progression use your real history; readiness lowers volume; health caps apply; the outdoor day, hydration breaks and time trimming work.

> **Suggested fix (small):** pick from the *top* N scored exercises of each pool (for example the best 8) and let the daily seed choose among those. That keeps rotation but makes recency, goal, readiness and the AI count. It is a contained change in one function and can be covered by tests like the one above.

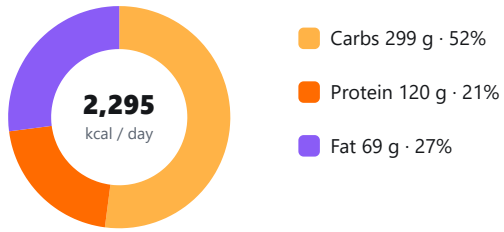
PART C · NUTRITION ALGORITHM

Step 1 — Daily targets and water

Before a single food is picked, the app works out how much you should eat and drink. This is plain arithmetic on your profile.

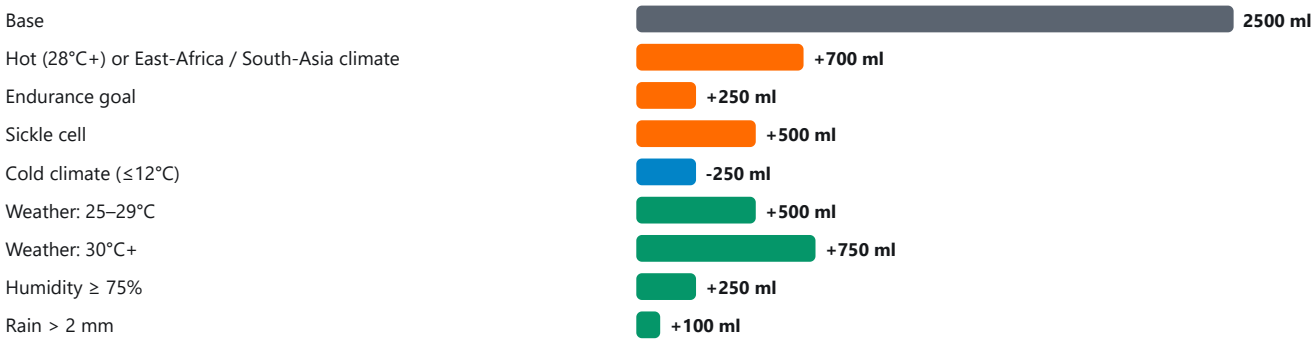
Calorie and macro targets (health/targets.ts)

1. If you typed your own calorie target, use it.
2. Otherwise, **Mifflin-St Jeor**: `10×kg + 6.25×cm − 5×age + 5` (male) / `− 161` (female) / `− 78` (unspecified).
3. × activity factor `min(1.55, 1.2 + 0.05 × training days)`.
4. Goal: fat loss −400, muscle gain +300; at least 1,200 kcal.
5. Protein = **1.6 g/kg** (2.0 for fat loss or muscle gain).
6. Fat = **27%** of calories ÷ 9. Carbs = the rest ÷ 4.



Worked example — male, 30 y, 75 kg, 175 cm, 3 training days, general fitness: base = 750 + 1,093.75 − 150 + 5 = 1,698.75 → × 1.35 = 2,293 → rounded to **2,295 kcal**. Protein 75 × 1.6 = **120 g**. Fat 2,295 × 0.27 ÷ 9 = **69 g**. Carbs (2,295 − 480 − 621) ÷ 4 = **299 g**. If height, weight or age is missing and you set no target, the app falls back to 2,200 kcal / 130 g protein.

Water target (ml per day)

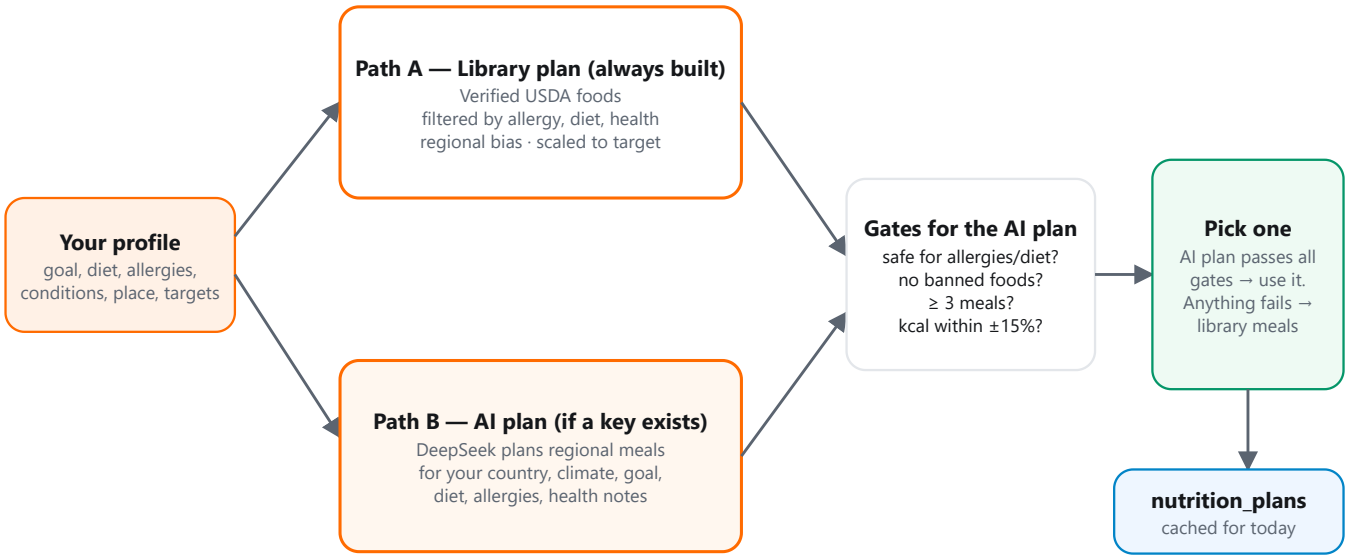


Open-Meteo gives the temperature, humidity and rain at your saved location. The total is clamped between 1,800 and 5,000 ml and shown as glasses of 300 ml. Without a location the base is used.

PART C · NUTRITION ALGORITHM

Step 2 — Two ways to plan, one safe result

The library plan is built first and is always ready. The AI plan is offered second and only used if it passes every gate.



Detail	How it works
Cache	One plan per person per day (table nutrition_plans). Reloading never re-calls the AI. The plan is rebuilt only if something that feeds it changes: goal, diet, allergies, calories, weight, height, meals/day, cultural rules, conditions, pregnancy, location or training days. The “New meal plan” button forces a rebuild
What the library always supplies	Hydration target, health notes, location label and the fallback meals — even when the AI wins
Which meals	Breakfast, lunch, dinner (+ a snack if you chose 4 meals) and a post-workout meal on days you have a workout plan
Why two paths	The library is reliable but limited to what USDA lists; the AI knows local dishes (ugali, injera, dal, tortilla…) the dataset lacks. Safety is guaranteed by running the same allergy / diet / health checks on both

PART C · NUTRITION ALGORITHM

Step 3 — The library algorithm in detail

How the app builds meals from real USDA foods (about 340 Foundation + 7,793 SR Legacy records).

#	Stage	What happens
1	Safe pool	Keep foods that are verified, have calories, are safe for your allergies (milk, peanut, tree nut, egg, soy, fish, shellfish, gluten and tag matches) and your diet (vegan / vegetarian), and do not match a health ban (salty, sugary, raw fish…)
2	Region	Country text → east_africa / south_asia / europe / americas / global. If at least 4 safe foods match that region's staples (e.g. plantain, cassava, ugali, teff, beans, lentils, rice, sweet potato for East Africa), they are moved to the front of the pool
3	Climate	Weather ≥ 28°C = hot, ≤ 12°C = cold; with no weather, East-Africa / South-Asia count as hot. Hot raises the water target
4	Pick by role	From the pool, take 3 protein foods (chicken, egg, yogurt, tuna, lentil, bean…), 3 carbs (rice, oat, potato, banana…), 3 vegetables and 2 fats (avocado, nuts, olive oil…). Position in each list = (daily seed + offset) mod list size, so it rotates daily
5	Assemble	Breakfast = protein + carb · Lunch = protein + carb + vegetable · Post-workout = protein + carb · Dinner = protein + vegetable + fat
6	Per-food numbers	Each food's calories/macros per 100 g × its serving grams ÷ 100
7	Scale to target	factor = target ÷ plan total, limited to 0.75–1.35, and applied to every food. Then meal totals are recomputed

Limits you should know. The scale factor is capped, so with an unusual target the day can still end below or above it. Foods are chosen by name patterns, not a nutritionist's rules, so combinations can be plain. This is exactly why the AI path exists.

Health rules in nutrition

The same condition table from page 7 supplies the food bans and notes (high blood pressure/heart/kidney → no salty foods; diabetes → no sugary foods; pregnancy → no raw fish, liver, high-mercury fish, alcohol). They are applied to the library pool *and* to every food name the AI proposes.

PART C · NUTRITION ALGORITHM

Step 4 — The AI plan, logging and photo check

How DeepSeek is asked, how its answer is policed, and how your meals are counted.

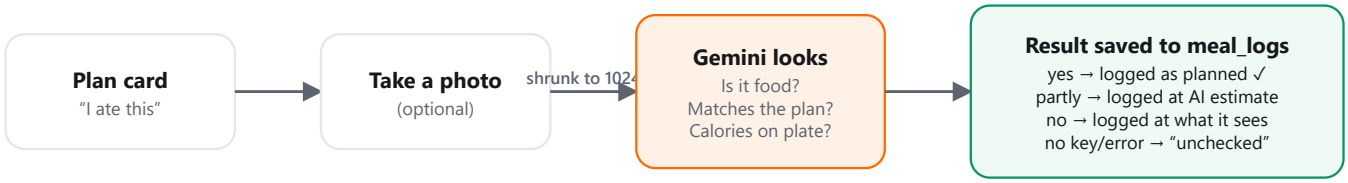
The question sent to DeepSeek

Sent	Details
Place & climate	City, country, hot / mild / cold
Goal & targets	Daily calories, protein, carbs, fat
Safety	Diet style, allergies, cultural / religious rules, health notes
Format	Exact slots in order, 2–5 foods per meal, portion + calories + macros per food, total within 5% of target, a day-seed so it varies
Role	"Registered-dietitian style planner… local, affordable foods… respect every allergy… no medical diagnoses". Temperature 0.6

How the answer is policed

1. Must be valid JSON with a meals list; numbers are clamped to sane ranges.
2. **Any** food that fails the allergy, diet or health-ban check cancels the whole AI plan (it is never quietly edited).
3. At least 3 meals, and the total must be between 85% and 115% of your calorie target.
4. If any gate fails the library meals are used. The screen says which source built today's plan: "✦ AI regional plan" or "Verified food library".

Logging a meal and the photo check



Honest limits. AI calorie and macro numbers are *estimates* and are labelled so. Photo checking needs a Gemini key; without one the meal is logged as planned and marked unchecked. Photos are shrunk and analysed, not stored. Meals you log feed the dashboard's calories, macros and nutrition score.

PART D · THE REST

Dashboard: where every number comes from

One call (/api/dashboard) reads your recorded data and returns the lot. When nothing is recorded the value is empty and the screen says so — no invented scores.

Number	Formula	Data used
BMI	kg ÷ (height in m)². <18.5 underweight · <25 healthy · <30 overweight · else obese	profile weight, height
Goal %	(current − first recorded weight) ÷ (target − first recorded weight), clamped 0–100	weight_history, target weight
Calories today	sum of calories of meals you logged vs your daily target	meal_logs
Macros today	sum of protein/carbs/fat of logged meals vs targets	meal_logs
Adherence (7 days)	days with a finished workout ÷ your weekly sessions, max 100%	workout_logs
Streak	consecutive days (counting back from today, or yesterday) with a finished workout	workout_logs
Fitness score	100 × (0.5 × 30-day consistency + 0.3 × average completion + 0.2 × BMI band). BMI band: healthy 1, overweight 0.6, obese 0.3, unknown 0.6. Empty until you finish a workout	workout_logs, BMI
Recovery	today's readiness, else average of the last 7 days	daily_checkins
Body fat label	ACE bands, by gender (e.g. men: <6 essential, <14 athlete, <18 fitness, <25 average)	profile
Weeks to goal	|target − weight| ÷ 0.4 kg per week, limited to 4–52	profile
Calories burned	5 MET × kg × hours for each finished block (duration, or sets × 50 s) plus tracked runs — an estimate	activity_data, run_activities

Health overview score (0–100)

Part	Excellent / Good / Fair / Poor	Score
Recovery	≥80 / ≥65 / ≥45 / lower	95 / 80 / 55 / 30
Sleep (7-day average hours)	≥7.5 / ≥7 / ≥6 / lower	95 / 80 / 55 / 30
Nutrition (7-day average kcal ÷ target, needs ≥3 logged days)	Good 85–115% · Fair 70–130% · else Poor	85 / 55 / 30
Hydration (7-day, needs ≥3 days)	≥90% / ≥70% / ≥50% / lower	95 / 80 / 55 / 30
Overtraining	Low / Moderate / High: sessions vs weekly plan, and sore (≥4) on top of extra sessions	95 / 60 / 30

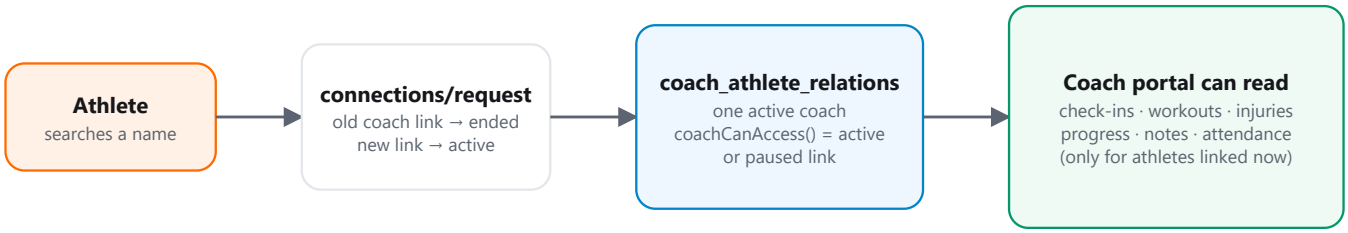
Overall = the average of the parts that have data (at least two). Parts with no data are shown as “No data” and a note says why (e.g. “Sparse meal logging“).

PART D · THE REST

Coach connection and payments

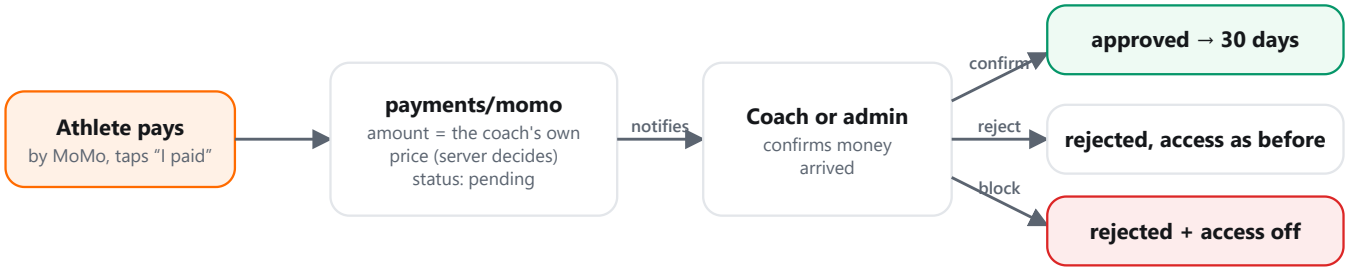
Data follows the athlete's current coach. Money follows the coach's own price.

Connecting



Leave or switch coach → the link ends and the old coach's access stops immediately. Admin can see all athletes and coaches.

Paying



Rule	Detail
Fixed price	A coach sets price (RWF per 30 days) and MoMo number in Settings. The payment API ignores any amount from the browser for coach memberships
30-day cap	Every approval or direct grant sets expiry to now + 30 days; renewing early never stacks days
Who can confirm	The coach who owns the order (and only for athletes still linked to them) or an admin. An answered order cannot be answered again
No coach	The athlete pays Kotaana directly (platform price, default $10 converted to RWF) and an admin confirms

PART D · THE REST

API map

Every route, what it needs and which engine/table it uses. “A” = athlete, “C” = coach, “Ad” = admin.

Route	Who	Does	Engine / tables
GET workout/generate	A	Today's session; auto-builds when missing or health changed	workout engine, ai rerank · workout_plans
POST workout/generate	A	Rebuild now (after the questions)	same
POST workout/perform	A	Save effort per block, update completion, add calories	exercise_performances, workout_logs, activity_data
GET exercises/[id]	A	Steps, demo media for “How to do it”	exercises
GET / POST nutrition/daily	A	Today's meals (cached) / force a new plan	nutrition engine, meal-plan AI, targets · nutrition_plans, foods
POST nutrition/log	A	Log or undo a meal, optional photo check	Gemini vision · meal_logs
GET / POST hydration	A	Glasses of water for your local day	hydration_logs
GET / POST health/checkin	A	Daily readiness card	daily_checkins
health/injury · health/document	A	Injuries and medical files that feed the safety filters	injuries, medical_documents
GET dashboard	A	All dashboard numbers	targets, many tables (read only)
reports/overview · reports/series · ai/weekly-summary · ai/monthly-progress · ai/risk	A	Recorded-data reports and risk signal	reports-data
GET connections/search	A	Find coaches by name (price, MoMo)	users, profiles
POST / DELETE connections/request	A	Choose a coach / leave	coach_athlete_relations, notifications
coach/members · coach/access · coach/notes · coach/follow-ups · coach/injuries · coach/sessions · coach/attendance	C	Manage members and coaching tools	relations + member data
PATCH coach/momo	C	Set price, MoMo number and name	profiles
POST payments/momo	A, C	Create a payment order (amount decided by server)	payment_requests, notifications
POST payments/momo/approve	C, Ad	Confirm / reject / block	payment_requests, subscriptions
payments/momo/webhook	MoMo	Provider status updates (signed)	payment_requests
admin/metrics · admin/subscriptions · owner/platform-momo · platform/momo	Ad	Totals, price, access, platform code	app_config, subscriptions
profile · location · notifications · progress/photo · track/run	A	Profile, GPS & weather place, inbox, photos, runs	profiles, run_activities…
auth/*	all	Sign-up, sign-in (password / Google), reset	users, accounts

PART D · THE REST

Database map and honest limits

Which tables feed which decision — and where the system is weaker than it looks.

Decision	Reads	Writes
Build workout	profiles, exercises, injuries, exercise_performances, workout_logs, daily_checkins, coach profiles (equipment)	workout_plans
Build nutrition	profiles, foods, workout_plans (post-workout meal), Open-Meteo	nutrition_plans, profiles (water target)
Log food / water	nutrition_plans, hydration_logs	meal_logs, hydration_logs
Dashboard	profiles, weight_history, daily_checkins, meal_logs, hydration_logs, workout_logs, activity_data, run_activities, injuries	—
Coach view	coach_athlete_relations + the athlete's data above	notes, follow-ups, attendance, sessions
Access & money	subscriptions, payment_requests, app_config	subscriptions, payment_requests, notifications

What to keep in mind

1. Exercise choice is a seeded draw from the safe pool (page 12). Scoring, recency and the AI's suggestion do not steer it yet.

2. Exercise data quality. All 1,002 records are rated "partial"; many lack instructions and need a human safety review. Filters depend on those tags.

3. The AI only sees the first 160 exercises in the table, not the safe pool, so even after the fix it can influence only part of the library.

4. Estimates are estimates: AI calories/macros, photo-based calories, calories burned (MET) and the fitness score formula are approximations and labelled as such.

5. Not medical advice. Condition rules are conservative filters and reminders, not clearance from a doctor.

6. Privacy by design: AI never decides safety, the AI request holds no name or email, and only the athlete's current coach (and admin) can read their data.

Quick reference — numbers worth remembering

Item	Value
Exercises in library	1,002 (3 sources)
Foods in library	≈ 8,130 (USDA Foundation 340 + SR Legacy 7,793)
Main blocks by time	≥50 min 5 · ≥38 4 · ≥28 3 · else 2
AI exercise bonus	+18 (not yet steering the pick)
AI meal plan accepted if	all foods safe, ≥3 meals, 85–115% of target kcal
Access length	30 days, never stacked
Water glass	300 ml; clamp 1,800–5,000 ml

CHK:ok